

Assignment 1: Dynamical Systems

Due Feb. 13 at 23:59

This assignment is to be done individually.

Important Note: The university policy on academic dishonesty (cheating) will be taken very seriously in this course. You may not provide or use any solution, in whole or in part, to or by another student.

You are encouraged to discuss the concepts involved in the questions with other students. If you are in doubt as to what constitutes acceptable discussion, please ask! Further, please take advantage of office hours offered by the instructor and the TA if you are having difficulties with this assignment.

DO NOT:

- Give/receive code or proofs to/from other students.
- Copy solution directly from Google or ChatGPT.

DO:

- Meet with other students to discuss assignment (it is best not to take any notes during such meetings, and to re-work assignment on your own).
 - Use online resources (e.g. Wikipedia) to understand the concepts needed to solve the assignment.
 - Indicate in a cover note how AI tools were used, if at all, to help with the assignment.
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Submitting Your Assignment

The assignment must be submitted on Canvas. Your submission must contain a zip file with the following files. These files must be in the root of the zip file.

- `report.pdf`: Written report
- `satellite.py`: Script containing RK4 implementation and simulation of a satellite orbiting Earth
- `dubins_flat_planner.py`: ROS2 node for differential flatness planning
- `dubins_flat_planning_algorithm.py`: Differential flatness planning algorithm backend
- `mcl.py`: ROS2 node for Monte Carlo Localization
- `mcl_algorithm.py`: Monte Carlo Localization algorithm backend

1 Satellite orbiting Earth

A simplified set of equations of motion for the Earth and an orbiting satellite is given by the following:

$$\ddot{r} = r\dot{\theta}^2 - \frac{k}{r^2} + u_1, \quad (1a)$$

$$\ddot{\theta} = -\frac{2\dot{r}\dot{\theta}}{r} + \frac{1}{r}u_2, \quad (1b)$$

where r represents the Earth-satellite distance measured from their centres, and θ represents the phase of the orbit. k is a positive constant.

- a) Derive a state space model of this system in the form of a first-order ordinary differential equation.
- b) What are the equilibrium points of the state space model, under zero control input, $u_1 = u_2 = 0$? Give a physical interpretation of the result.
- c) What are the equilibrium points of the state space model, under $u_1 = k/r^2, u_2 = 0$? Give a physical interpretation of this control set point and of the equilibrium points.
- d) Linearize the model with respect to a reference orbit given by $r(t) \equiv \rho, \theta(t) = \omega t, u_1 = u_2 = 0$.
- e) Numerically integrate the ODE, with $u_1(t), u_2(t) \equiv 0$, using your own implementation of RK4. Plot the state trajectory and intuitively explain the behaviour. Use parameters for the international space station orbiting the Earth:
 - $r(0) = 410 \text{ km} + 6378 \text{ km}$, which represents an orbit 410 km above the earth's surface,
 - $\dot{r}(0) = 0 \text{ m/s}$,
 - $\theta(0) = 0$,
 - $\dot{\theta}(0) = 2\pi/T$, where $T = 92.68$ minutes, the orbital period
 - $k = GM$, where $G = 6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$, $M = 5.97 \times 10^{24} \text{ kg}$.

See the readme of the starter code in `satellite.zip` to get started.

2 Open-loop planning with car-like model

Consider the extended Dubins car model:

$$\dot{x} = v \cos \theta \quad (2a)$$

$$\dot{y} = v \sin \theta \quad (2b)$$

$$\dot{\theta} = \omega \quad (2c)$$

$$\dot{v} = a \quad (2d)$$

The state consist of the position (x, y) , the heading θ , and the longitudinal speed v . The controls are the turn rate ω and the longitudinal acceleration a .

- a) Show that the system is differentially flat by letting $z = (x, y)$, and deriving the functions β and γ such that

$$(x, y, \theta, v) = \beta(z, \dot{z}, \dots, z^{(q)}) \quad (3a)$$

$$(\omega, a) = \gamma(z, \dot{z}, \dots, z^{(q)}) \quad (3b)$$

- b) Consider a maneuver given by the following initial condition at $t = 0$ and final condition at $t = T = 10$:

$$\begin{bmatrix} x(0) \\ y(0) \\ \theta(0) \\ v(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} x(T) \\ y(T) \\ \theta(T) \\ v(T) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{\pi}{2} \\ 1 \end{bmatrix} \quad (4)$$

Using the basis functions $\psi_0(t) = 1, \psi_1(t) = t, \psi_2(t) = t^2, \psi_3(t) = t^3$, we parameterize the flat outputs as follows:

$$x(t) = \sum_{i=0}^3 b_{0i} \psi_i(t) \quad (5a)$$

$$y(t) = \sum_{i=0}^3 b_{1i} \psi_i(t) \quad (5b)$$

Write a system of equations in terms of the coefficients $\{b_{0i}\}$ and $\{b_{1i}\}$ such that the initial and final conditions are satisfied.

- c) Solve the above systems of equations to obtain and plot the state and control trajectories. See `readme.md` in `dubins_flat_planner.zip` for detailed instructions.
- d) Integrate the above code into a ROS2 front end, and simulate open-loop planning in a ROS2 package. See `readme.md` in `dubins_flat_planner.zip` for detailed instructions.

3 Monte Carlo Localization

Consider the following simple discrete-time model of the Dubins car:

$$x_{t+1} = x_t + (v_t + \nu_t) \cos \theta_t \Delta t \quad (6a)$$

$$y_{t+1} = y_t + (v_t + \nu_t) \sin \theta_t \Delta t \quad (6b)$$

$$\theta_{t+1} = \theta_t + (\omega_t + w_t) \Delta t \quad (6c)$$

The state consist of the position (x, y) , the heading θ . The controls are the longitudinal speed v and the turn rate ω . At every time step, the system experiences control noises $\nu_t \sim \mathcal{N}(0, \sigma_v)$ on v and $w_t \sim \mathcal{N}(0, \sigma_\omega)$ on ω .

- a) Complete a basic implementation of map-based localization, Monte Carlo Localization (MCL). See `readme.md` in `mcl.zip` for detailed instructions.

Hints:

- σ_v and σ_ω are specified by the parameters `motion_noise_trans` and `motion_noise_rot` respectively
- The parameters `particle_noise_trans` and `particle_noise_rot` should be used to add post-resampling jitter to (x, y) and θ respectively
- Measurement update is skipped when the translational or angular displacement is less than The parameters `update_dist_thr` and `update_angle_thr`. (This has already been implemented.)